



Ruhila Goswami

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"Nothing exists for itself alone, but only in relation to other forms of life." – Charles Darwin

Research Profile

Focus: Evolutionary divergence and dental variation in salmonids.

Interests: Population genetics; phenotypic-genotypic mapping; structural variation in genomes and selection; salmonid evolution; non-invasive methods.

Methods: Phylogenetic inference, genomic synteny, PCR-based sex determination, skeletal phenotyping, field sampling, and reproducible analysis in R and Python.

Education

University of Iceland

2024–2026

M.S. Biological Science, Reykjavík, Iceland

Advisor: Prof. Arnar Pálsson. THESIS: Evolutionary Divergence in Dental Variation and Sex Chromosome Architecture in Icelandic Brown Trout and Arctic Charr.

IISER Mohali

2021–2024

B.S. Biology (Major), India

First Division with Honors.

Research Experience

University of Iceland

Summer 2023

Research Staff with Prof. Arnar Pálsson

Studied synteny of histone genes in salmonids, focusing on gene arrangement and neighboring genomic regions.

University of Iceland

Summer 2022

Research Staff with Prof. Arnar Pálsson

Reproducible phylogenetic workflow for possible molecular evolutionary pathways in salmonids: NCBI data curation, BLAST homology inference, MUSCLE5 alignments, Gblocks trimming, and BIONJ/RAXML-NG tree simulation. PROJECT REPORT: Computational Primitives for Evolutionary Paths ($\approx$ 147 pages).

IIT Kanpur

Winter 2022

Research Interns with Prof. Debabrata Goswami and Dr. Nagma Parveen

Studied non-linear optics, optical tweezers, viral wet-lab methods, and a SARS-CoV-2 pseudovirus protocol using an HIV-1 lentiviral packaging system.

IISER Mohali

Summer 2021

Research Intern with Dr. Lolitika Mandal

Explored genetic tools for *Drosophila* from a wet-lab perspective of data collection and analysis.

Publications, Posters, and Awards

Manuscript in preparation

2026

Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids.

Posters

2025–2026

Icelandic Ecological Society (Vistís), NOWPAS, and IceBio posters on tooth numbers and bone shape in Icelandic salmonids; Best Poster Award at Vistís 2025.

Selected proceedings and presentations

2022

IEEE PDGC proceedings on reproducible literate phylogenetic analysis and high-performance computing workflows; BES poster on *Salmo salar* histone sequence data.

Teaching

LÍF401G Developmental Biology

Spring 2026

Teaching Assistant, University of Iceland. Course led by Prof. Adam Ray Smith; assisted with embryo analysis.

LÍF541G Plant Physiology

Fall 2025

Teaching Assistant, University of Iceland. Course led by Prof. Victor Anthony Albert; led student discussions in a course focused primarily on plant genetics.

LÍF403G Evolutionary Biology

Spring 2025

Teaching Assistant, University of Iceland; undergraduate course on evolutionary theory, natural selection, adaptation, and the history of life.

Research Skills

Genomics: Population genetics, genomic synteny, structural variation, PCR-based sex determination, non-invasive sampling.

Phenotypes and fieldwork: Dental and craniofacial variation, bone extraction, otolith extraction, fish dissection, electrofishing, salmonid sampling and preservation.

Evolutionary methods: Phylogenetic tree construction (distance, maximum likelihood, Bayesian), BLAST, MUSCLE5, Gblocks, RAxML-NG, BEAST2.

Computing and lab: R, Python, shell, Git, Snakemake, reproducible literate programming, DNA extraction and preservation, PCR, microscopy, *Drosophila* tissue dissection, cell culture.

Service, Open Source, and Affiliations

Google Summer of Code

2023–2024

Student Developer on Python bindings and documentation for d-SEAMS/pySEAMS, including installation refinements, usability improvements, and repository maintenance.

IEEE P3173

2022–present

Secretary, IEEE Standards Committee for Endocrine Disrupting Chemical Hazard Labeling.

Memberships: Nordic Society Oikos; Icelandic Ecological Society (Vistís); Icelandic Biological Society (Líffræðigáttin); British Ecological Society; Genetics Society, UK; Genetics Society of America.